**Listing 4.7. Defining the loss function**

```
def loss_fn(preds, r):                                  1
    return -1 * torch.sum(r * torch.log(preds))         2
```

- ***1*** The loss function expects an array of action probabilities for the actions that were taken and the discounted rewards.
- ***2*** It computes the log of the probabilities, multiplies by the discounted rewards, sums them all, and flips the sign.

### 4.4.4. The full training loop

Initialize, collect experiences, calculate the loss from those experiences, backpropagate, and repeat. The following listing defines the full training loop of our REINFORCE agent.

**Listing 4.8. The REINFORCE training loop**

```
MAX_DUR = 200
MAX_EPISODES = 500
gamma = 0.99
score = []
1
for episode in range(MAX_EPISODES):
        curr_state = env.reset()
        done = False
        transitions = []
2

        for t in range(MAX_DUR):
3
        act_prob = model(torch.from_numpy(curr_state).float())
4
        action = np.random.choice(np.array([0,1]),
p=act_prob.data.numpy())5
        prev_state = curr_state
        curr_state, _, done, info = env.step(action)
6
        transitions.append((prev_state, action, t+1))
7
        if done:
8
            break

        ep_len = len(transitions)
9
```